

JC4004 – Computational Intelligence

Sessions 37 & 38: CLIP

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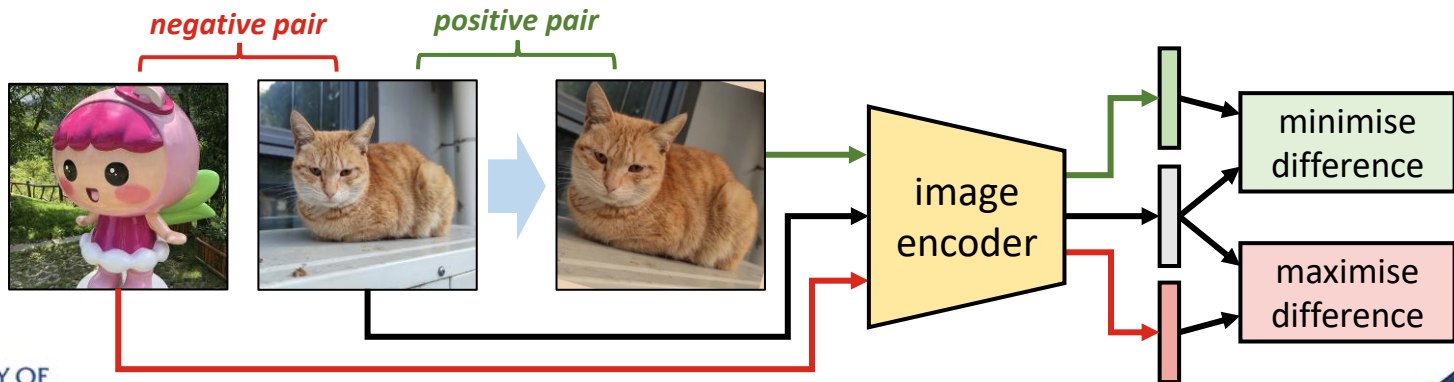
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What is CLIP?

- *Contrastive language-image pre-training (CLIP)* is a method for training models using images paired with corresponding text
 - Published by OpenAI (**Radford** et al.) in 2021
 - It is worth noting that CLIP is *not* a model but a training paradigm!
 - The goal of CLIP is that image and text models can produce embeddings in a common space (compatible and comparable with each other)
- CLIP is considered as an essential step towards general image-text models suitable for many tasks without fine-tuning
 - Models trained with CLIP have shown good performance in zero-shot image classification on established benchmarks such as ImageNet

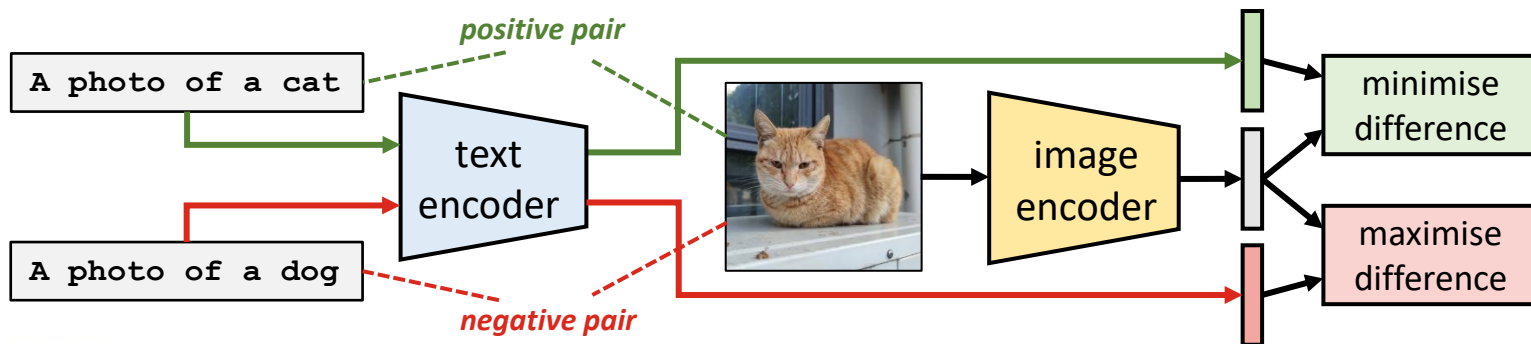
Contrastive learning fundamentals

- *Contrastive learning* minimises differences between positive pairs and maximises differences between negative pairs in the *embedding space*
 - Embeddings are the feature vectors from the model backbone (encoder)
 - Positive pairs are pairs of semantically similar items, whereas negative pairs are pairs of semantically different items (different classes)



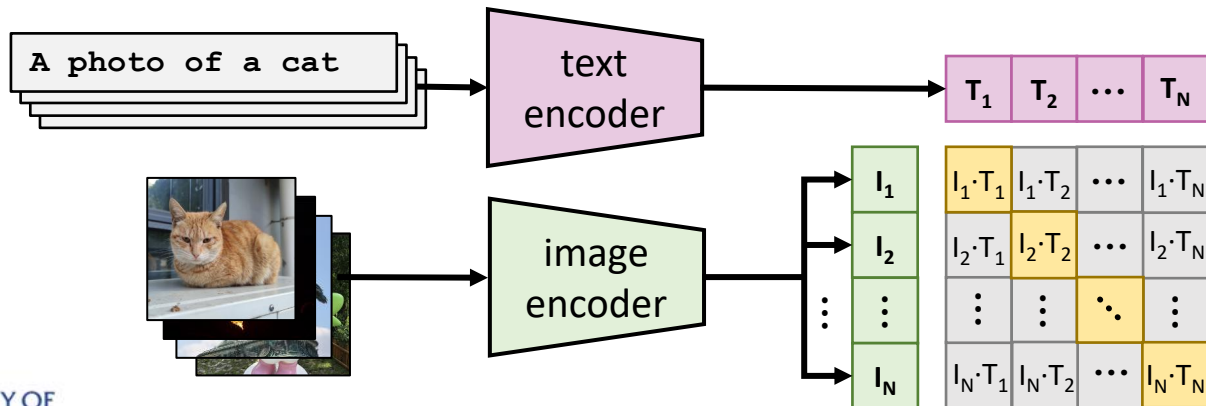
Contrastive learning for image-text pairs

- Classic contrastive learning uses pairs of same modality (e.g., images)
 - In CLIP, the idea is to use image-text pairs for contrastive learning
 - Cross-modal contrastive learning is a bit tricky, since the modalities need to share the embedding space even though they use different encoders
 - Encoders must be trained jointly, or one encoder adapted to the other



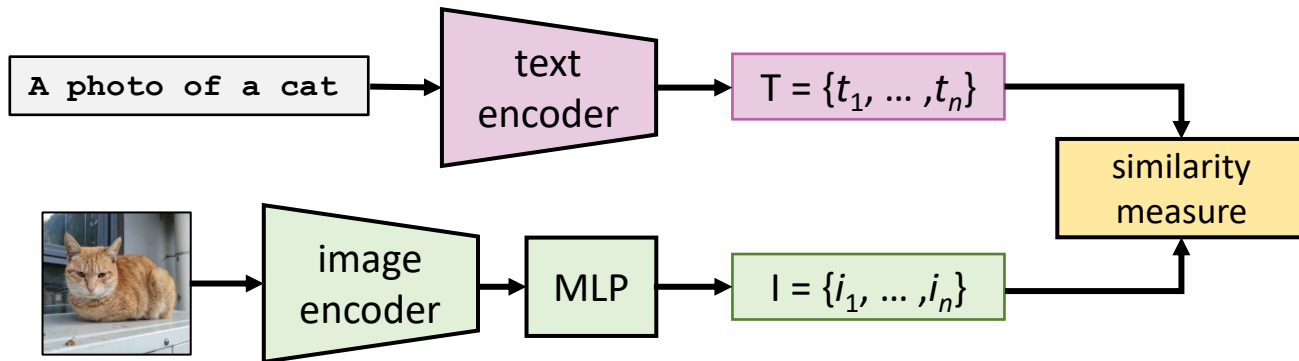
Contrastive pre-training in CLIP

- In contrastive pre-training in CLIP, we can start with N images and N related paired texts (e.g., image captions)
 - The goal is to predict which of the $N \times N$ possible pairings is correct
 - Basically, we can use any text encoder (e.g., BERT or GPT) and image encoder (e.g., ResNet or ViT) to generate embeddings



Pre-training fundamentals

- In CLIP, text and image encoders can be trained jointly from scratch
 - Typically, *cosine similarity* is used to measure similarity of embeddings
 - Given $N \times N$ image-text pairs, the objective is to maximise the cosine similarity of the N correct pairs and minimise the cosine similarity of the $N^2 - N$ incorrect pairs



Cosine similarity

- Cosine similarity is defined as cosine of angle θ between two non-zero vectors $\mathbf{a} = \{a_1, \dots, a_n\}$ and $\mathbf{b} = \{b_1, \dots, b_n\}$ of same length n
- We can derive cosine from Euclidean dot product formula:

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}}$$

- The resulting similarity range is $[-1,1]$, where -1 means the vectors are exactly the opposite and 1 means they are exactly the same

CLIP loss function

- In CLIP, each minibatch includes $N \times N$ image-text pairs
 - Total loss L for the minibatch can be computed by symmetric cross-entropy across the rows and columns (t is a learnable parameter):

$$L = -\frac{1}{2N} \left(\sum_{i=1}^N \ln \frac{\exp(I_i \cdot T_i/t)}{\sum_{j=1}^N \exp(I_i \cdot T_j/t)} + \sum_{j=1}^N \ln \frac{\exp(I_j \cdot T_j/t)}{\sum_{i=1}^N \exp(I_i \cdot T_j/t)} \right)$$

- The embeddings are normalised to unit vectors, so dot products represent cosine similarity directly
- Note that ground truth similarity is 1 when $i = j$, and 0 when $i \neq j$; therefore, we only need to include diagonal positions in summation

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Image-text SoftMax

similarity directly

Text-image SoftMax

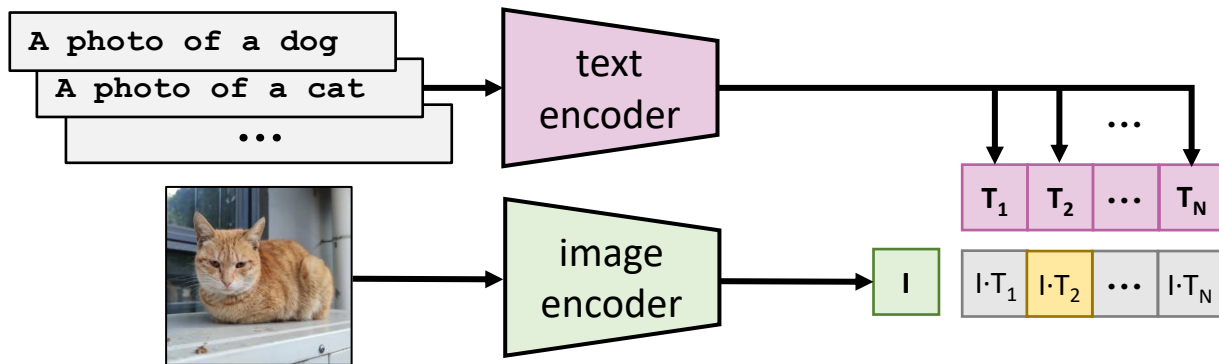
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Training data collection

- Labelling of datasets for supervised learning is slow and expensive
 - For example, the full ImageNet dataset with 14 million annotated images required 25,000 workers for labelling
- CLIP does not require labels representing a set of predefined classes
 - Can be trained on image-text pairs publicly available on the Internet
 - To cover a broad set of visual data, the original CLIP dataset with 400M pairs was collected using 500k queries, up to 20k results per query
 - The queries include all the words occurring at least 100 times in the English version of Wikipedia

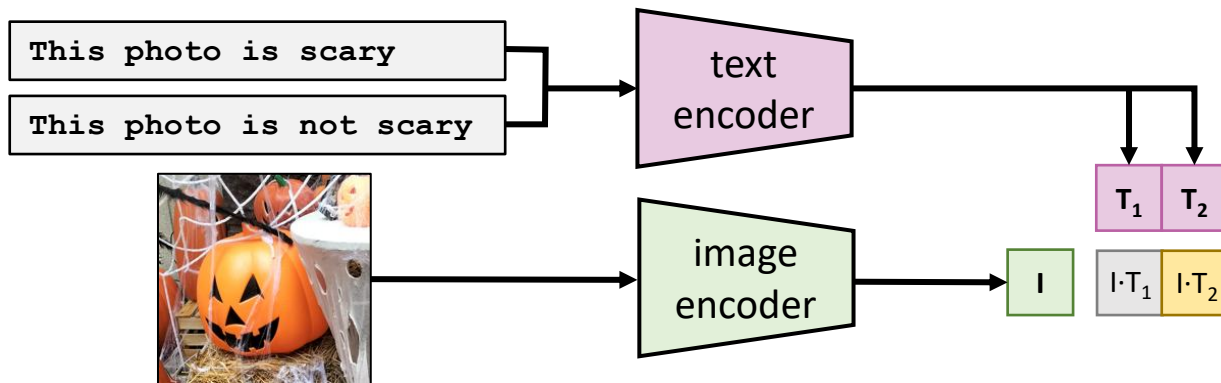
CLIP inference for classification

- A properly trained CLIP model can be used to find the text among several candidates that matches best with the input image
 - For example, we can use CLIP model for classification by encoding a set of sentences like “*Image of X*” where X describes the class (e.g., “*a cat*”)
 - The image-text pair with the highest similarity determines the image class



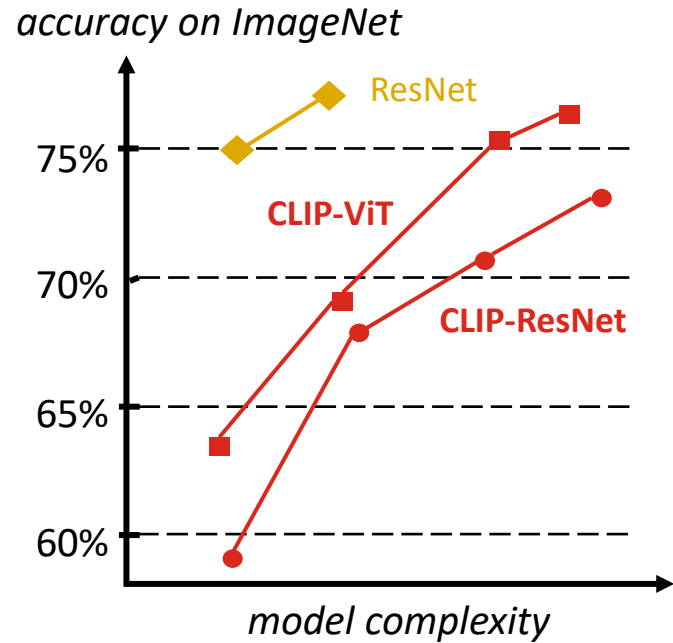
CLIP inference for other tasks

- By comparing against different input texts (prompts), CLIP can be used for tasks beyond classical object recognition
 - For example, we could test qualitative attributes of an image using prompts such as *“This image is beautiful”* or *“This image is blurry”*



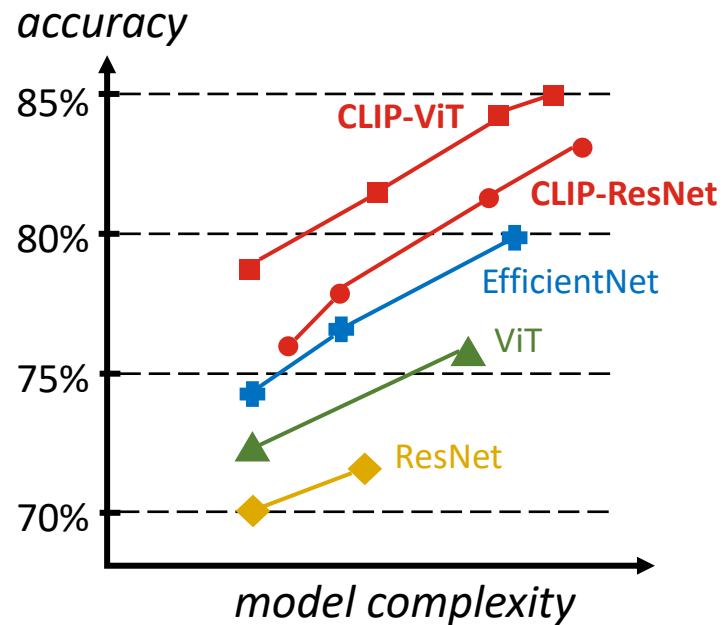
Zero-shot performance of CLIP

- Several ResNet and ViT variants trained using CLIP, GPT-2 used as language model
- CLIP models perform well in image classification without fine-tuning (zero-shot), but they still fall behind fine-tuned models
 - However, the main benefit of CLIP is its ability to generalise to unseen domains and datasets



Linear probing performance of CLIP

- Dataset-specific prediction head trained for each experiment (*linear probing*)
 - Results indicate that CLIP models generalise better than prior models pretrained on ImageNet
 - At the time of publishing, CLIP achieved state-of-the-art average results on 27 benchmark datasets



Limitations of CLIP

- CLIP performs well on recognising common objects, but struggles with objects not widely covered in the pre-training dataset
 - For example: fine-grained classification, like differentiating models of cars
 - Fine-tuning or linear probing still needed for specialised tasks!
- CLIP performs poorly on systematic and abstract tasks
 - For example: counting objects or estimating distances between objects
- CLIP results can be sensitive to the formulation of the prompt
- CLIP models are *not* generative models: they can *compare* images and text, but they cannot be used for e.g., generating image captions

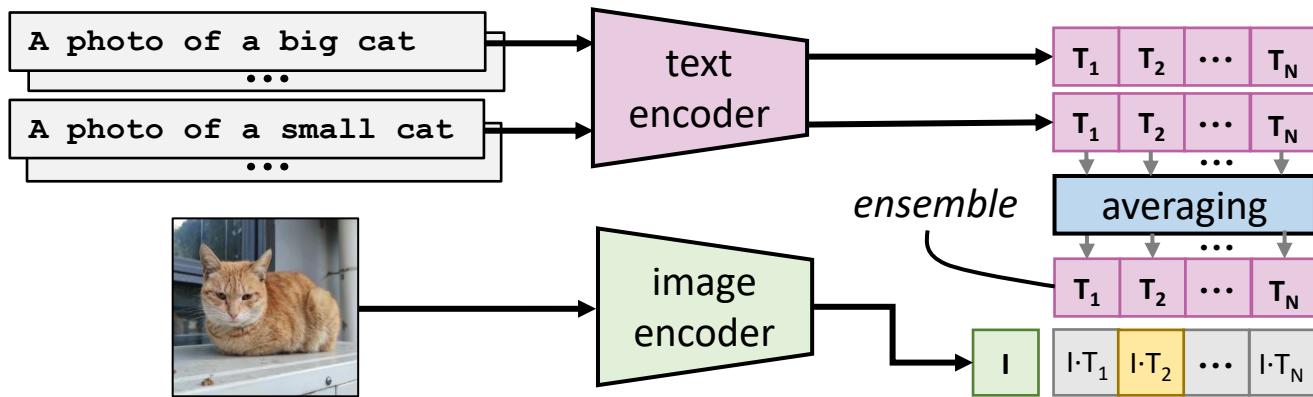
Break: any questions?

Prompt engineering

- A significant weakness of CLIP is that the results are sensitive to the formulation of the prompt
 - We can use different sentences with the same meaning, e.g., *“A photo of a cat”* or *“This image shows a cat”*, and the results will be different
- There are no clear rules for phrasing the prompts, so the proper phrasing is mostly learned by trial and error
 - Zero-shot performance on specific datasets can be improved by customising the prompts for the dataset
 - For example, performance on a satellite image dataset can be improved by using the phrase *“A satellite photo of a {label}”*

Prompt ensembling

- Performance of CLIP can be improved by using several prompts with different phrasings and ensembling them in the embedding space
 - For example, it is possible to ensemble different contexts represented by sentences like *“A photo of a big cat”* and *“A photo of a small cat”*



Sigmoid loss for CLIP

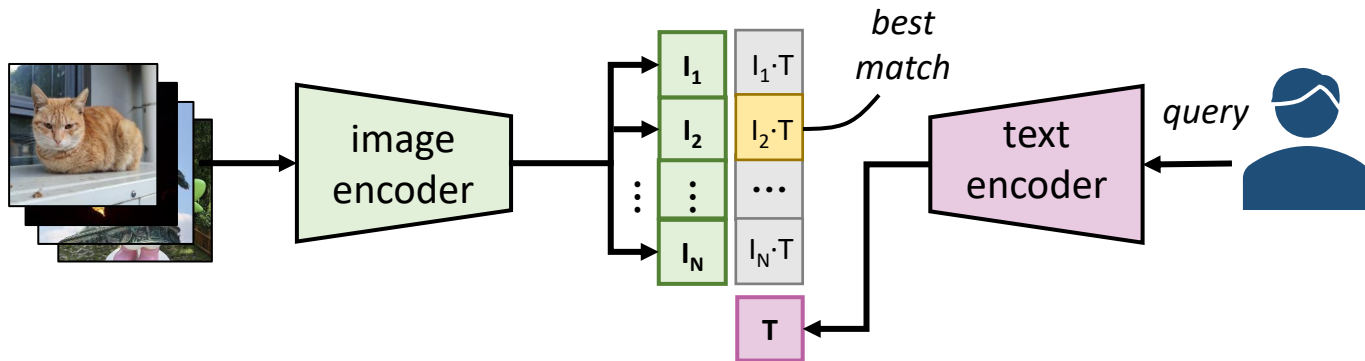
- **Zhai et al. (2023)** proposed to replace SoftMax by Sigmoid loss:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \log \frac{1}{1 + \exp(z_{i,j}(-t l_i \cdot T_i + b))}$$

- $z_{i,j}$ is set to 1 for positive image-text pairs and -1 for negative pairs
- Parameter $t = e^\tau$; τ and b are learnable parameters initialised to $\tau = 10$ and $b = -10$, respectively
- Sigmoid loss improves model accuracy slightly and it is more efficient to compute, especially in parallel processing architectures

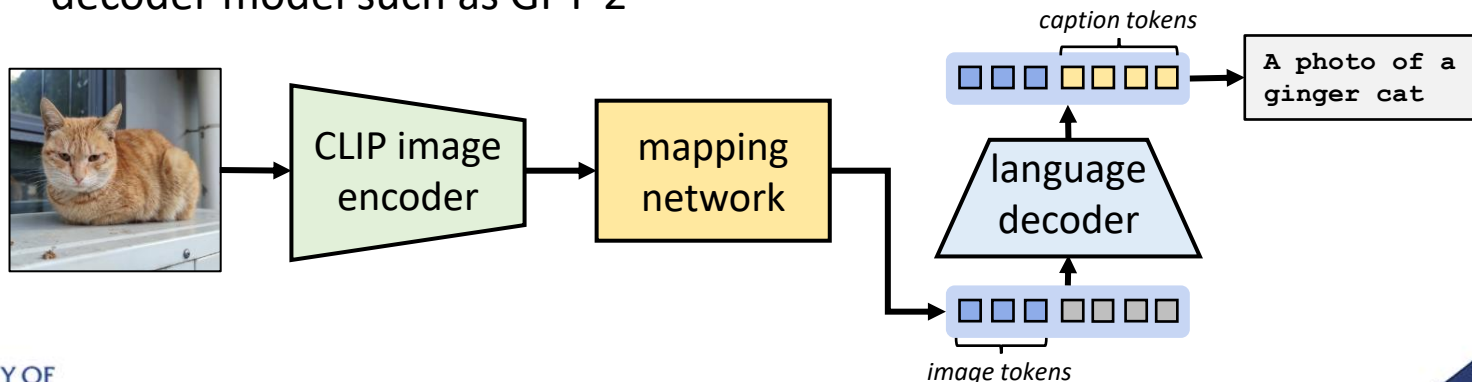
CLIP for text-based image search

- It is possible to store all the CLIP image embeddings for an image database in an index table and use it for text-based search
 - Image descriptions written by users are encoded, and the text embeddings are compared to the image embeddings to find the best matching image



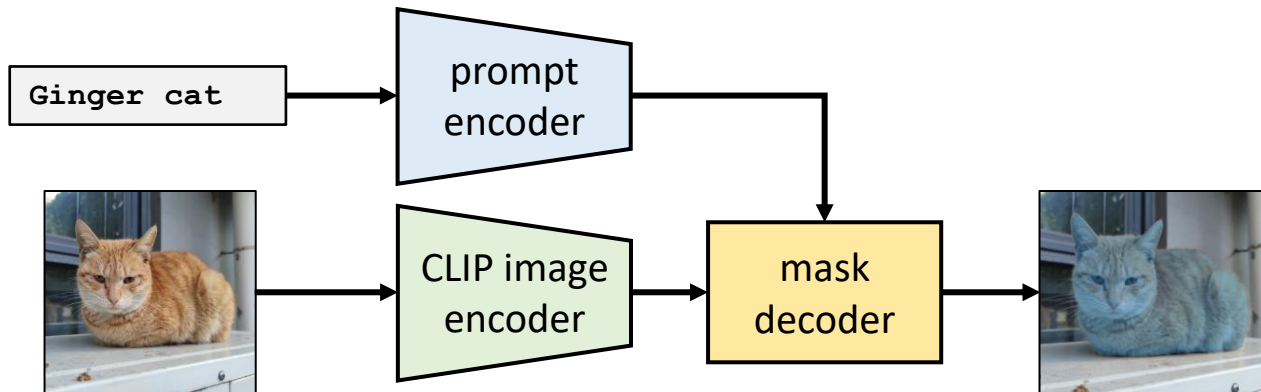
CLIP for image captioning

- Although CLIP models are not a generative model themselves, CLIP encoders have been adopted to image captioning systems
 - Frozen CLIP encoder used as input to a separate mapping network trained to transform CLIP embeddings to a language model embedding space
 - Tokens derived from CLIP image embeddings used as input to a language decoder model such as GPT-2



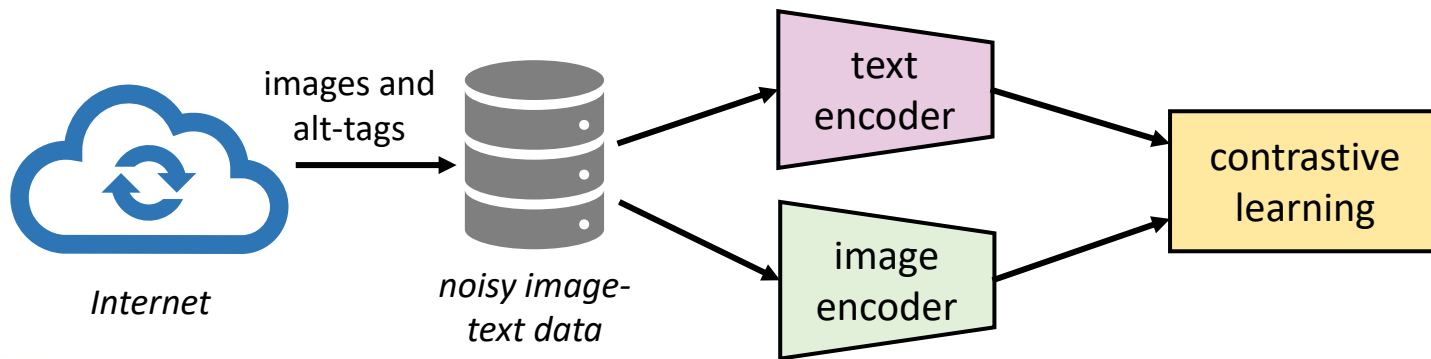
CLIP for image segmentation

- Many image segmentation models, including the prominent *segment anything model (SAM)*, include a CLIP image encoder
 - Image embeddings and prompt embeddings are used jointly as input to a lightweight mask decoder that generates the segmentation masks



ALIGN

- A large-scale image and noisy text embedding (ALIGN) is based on the same idea as CLIP, but using a larger (not much filtered) dataset
 - Data collected by using images and their alt-tags from the Internet: noisier than the CLIP dataset, but even larger with 1.8B image-text pairs
 - BERT used as text model instead of GPT-2

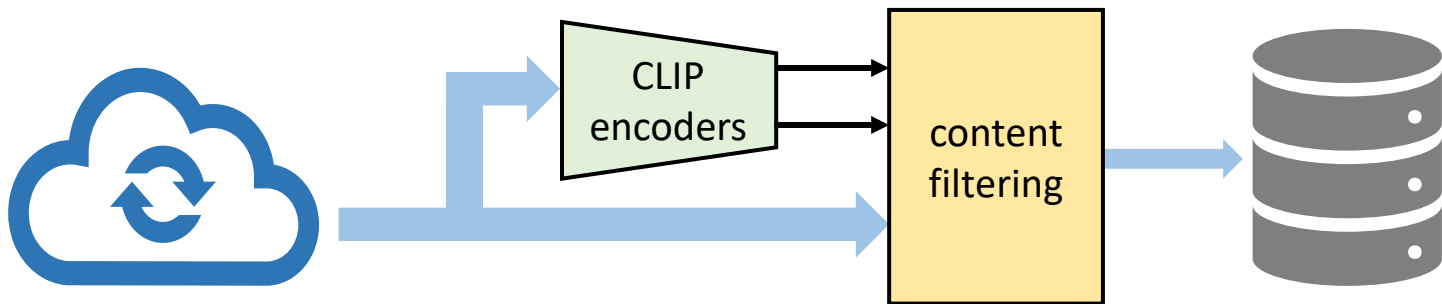


LAION public dataset

- Even though pre-trained CLIP models have been published, the original datasets for training CLIP and ALIGN are private
- *Large scale artificial intelligence open network (LAION)* constructed a large public dataset with image-text pairs in 2022
 - LAION-5B dataset with 5.85 billion image-text pairs, including LAION-2B subset that contains 2.32 billion pairs in English language
 - Performance of the CLIP models trained from scratch using LAION-400M subset of LAION-5B is similar with the original CLIP models

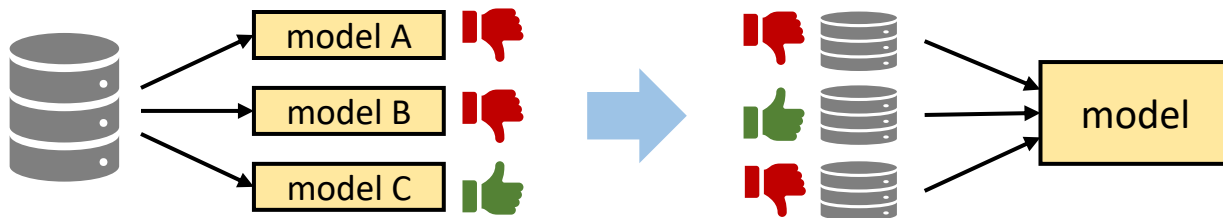
LAION dataset construction

- LAION dataset is collected by crawling for images with alt-text tags
 - Common Crawl metadata used to identify images with alt-text
 - The final dataset is *CLIP-filtered*: image-text pairs indicating too low similarity by the original CLIP models were discarded
 - CLIP embeddings also used to identify and remove illegal content



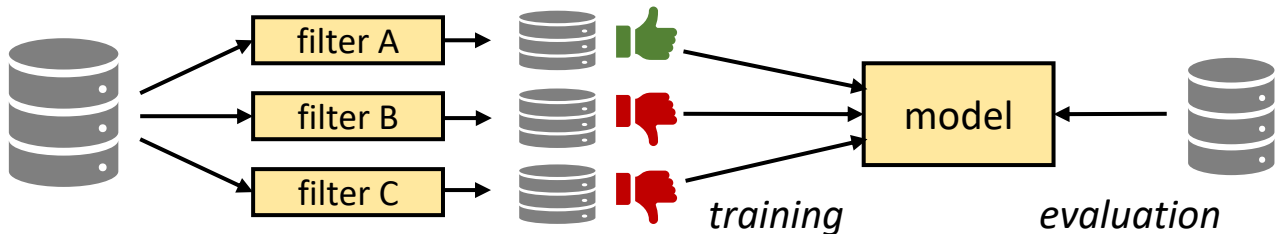
DataComp challenge

- Traditionally, AI research has mostly focused on developing model architectures that work well on established benchmark datasets
 - But what if the model architectures are good enough already, and the performance bottleneck is the datasets?
- DataComp challenge tries to shift focus from models to datasets
 - Instead of developing models and testing them on fixed datasets, in DataComp, new datasets are developed and tested on fixed models



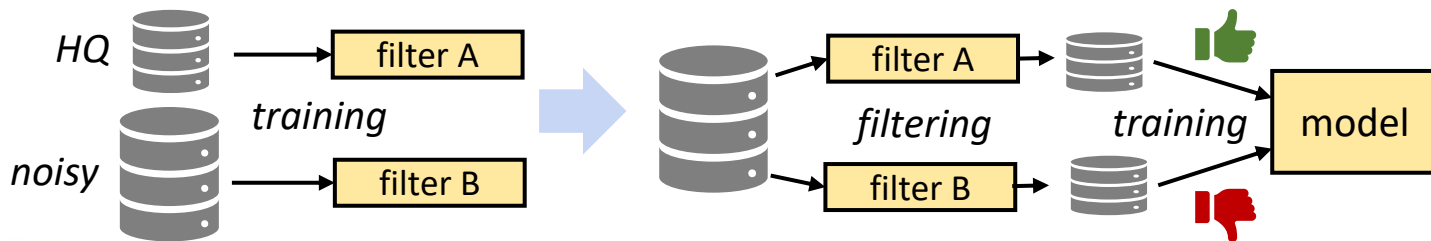
DataComp benchmark

- In DataComp challenge, participants contributed by either developing their data filtering techniques or suggesting data sources
 - Proposed data filtering techniques applied to a massive *CommonPool* dataset of 12.8 billion image-text pairs from Common Crawl
 - Filtered datasets used for training CLIP models and evaluated for their zero-shot performance on 38 different tasks
 - As a result, a high-quality benchmark dataset DataComp-1B was released



Data filtering networks

- Both LAION-2B and DataComp-1B datasets have been constructed by using CLIP models for filtering the content
 - CLIP models typically trained on large but noisy datasets: however, a data filtering model trained on a smaller but cleaner dataset can work better
 - DFN dataset was curated by using a data filtering model trained on a dataset with human verified image captions
 - CLIP models trained on DFN-2B achieved state-of-the-art accuracy



CLIP status and trends

- CLIP models are widely used in different applications, either alone or as a component of a larger system
 - Some new “CLIP-friendly” model architectures have been proposed, e.g., ViTamin, but the most prominent CLIP models are still based on ViT
 - The research focus have shifted towards development of better datasets
- OpenCLIP is an open source version of CLIP
 - Publicly available in https://github.com/mlfoundations/open_clip
 - Includes several CLIP models of different sizes, pre-trained on public datasets such as LAION-2B and DataComp-1B

Summary

- CLIP training paradigm uses contrastive loss between image-text pairs
 - The goal is to train image and text encoders to compare similarity of image-text pair using the embeddings
 - Image-text pairs readily available on the Internet: massive datasets for training CLIP models can be created without expensive manual labelling
 - CLIP models are useful for zero-shot image classification (no fine-tuning)
- CLIP is widely adopted to many applications and still evolving
 - CLIP models used as a component in several models developed for different applications like image captioning and segmentation
 - New larger datasets for training CLIP models are still under development

References

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Questions and discussion